

Experimental Reflections

Before this experiment, I thought machine learning was mainly about using prebuilt libraries to process data. However, this experiment helped me understand that the complete six step workflow is a process of testing, making mistakes, and finding answers. When working with the Wine dataset, I learned more about the "garbage in, garbage out" principle. Feature selection is not random. For example, when we tried different feature combinations, such as using the first four dimensions or choosing different chemical components, the model accuracy changed a lot. This helped me understand that feature engineering is not only a technical task. It also requires us to understand the data and relationships between different features.

When I compared logistic regression and decision trees, logistic regression had a higher accuracy of 88.9%. At first, I was surprised because I thought the decision tree was a more complex model and should perform better. After looking at the misclassifications in the confusion matrix, I understood the reason better. When there are only a few features and the decision boundaries are more linear, simple models such as logistic regression can work better. They can also be less likely to overfit. This changed my previous idea that a more complex model is always better.

This experiment helped me understand how a model performs when it works with unseen data. In the future, I want to use these workflows with more complex and noisy real-world data. I also want to learn more about cross validation and hyperparameter tuning. These methods can help reduce overfitting and improve model performance.